



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

SECV 2223

WEB PROGRAMMING

ASSIGNMENT 2: PORTFOLIO

REFLECTION

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SECTION : 04

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For this assignment, my initial design process was to establish a modern-minimalistic look that remained consistent throughout all the pages. After a few trials and errors, I eventually settled on one of my favourite colour combinations, black and blue.

Next, I decided to create a total of four pages, the 'Home' page, the 'About Me' page, the 'Projects' page and the 'Contact Me' page. Each page also consists of 4 main elements, the navigation bar, header, main content and footer. The navigation bar was to allow for easy navigation throughout the four pages, whereas the header would introduce briefly the page's content with a big title and supporting subtitle with more details. The main content section would house the main purpose of the page and the footer containing my name, matric number, course code, name and assignment number.

To make the pages more visually appealing, I applied CSS and JavaScript concepts that I've learnt so far in my course, including a dynamic greeting that checks the time of day and greets the user with a 'Good Morning, Afternoon or Evening' based on the time. I also added a project filter with JavaScript, loops and DOM manipulation so that projects can be filtered by category. Additionally, media queries were added to cater to proper navigation on different devices with different screen sizes.

After the draft webpages were coded, I conducted a Google Lighthouse audit and initially did not score 100s across the board. The worst score was Performance, where I scored 73 which was mainly due to my usage of images in the .png format. To improve these scores, I converted and compressed the images used to a more modern .webp format to improve load speeds. Additionally, in some page the cumulative layout shift (CLS) was affecting the scores due to page movements. To combat this, I added fixed grid sizing, minimum heights to ensure the page does not shift too much during initial load on affected pages. Next was Accessibility, whereby some links were not distinguishable. I solved this by improving contrast and adding active and hover animations. As for Best Practices, Lighthouse detected browser errors logged into the console as a result of the `getGreeting()` function being referenced on pages that do not use it. This was solved by adding null checking in the JavaScript for webpages that do not use the `getGreeting()` function. Finally, for SEO, I initially scored 82 because all my pages did not contain a meta description, which affected the scores. Then, I promptly added the appropriate meta description for each page which fixed this issue.